

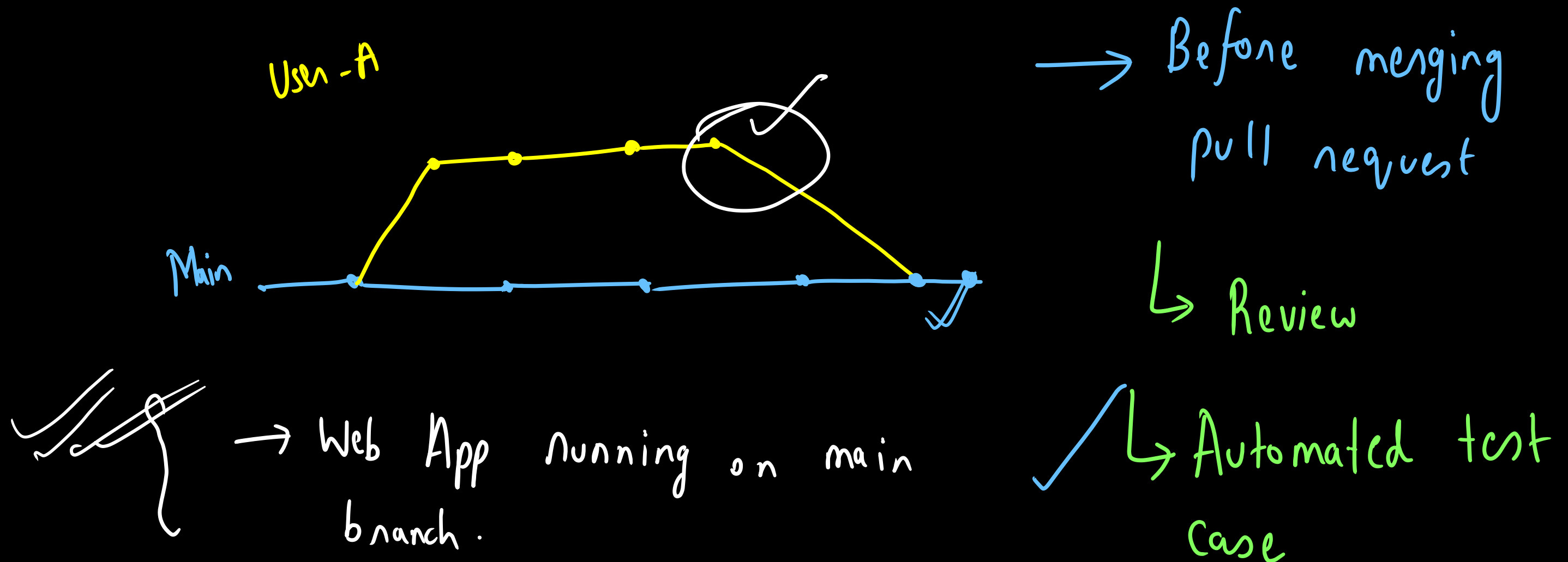
Continuous Integration (CI)

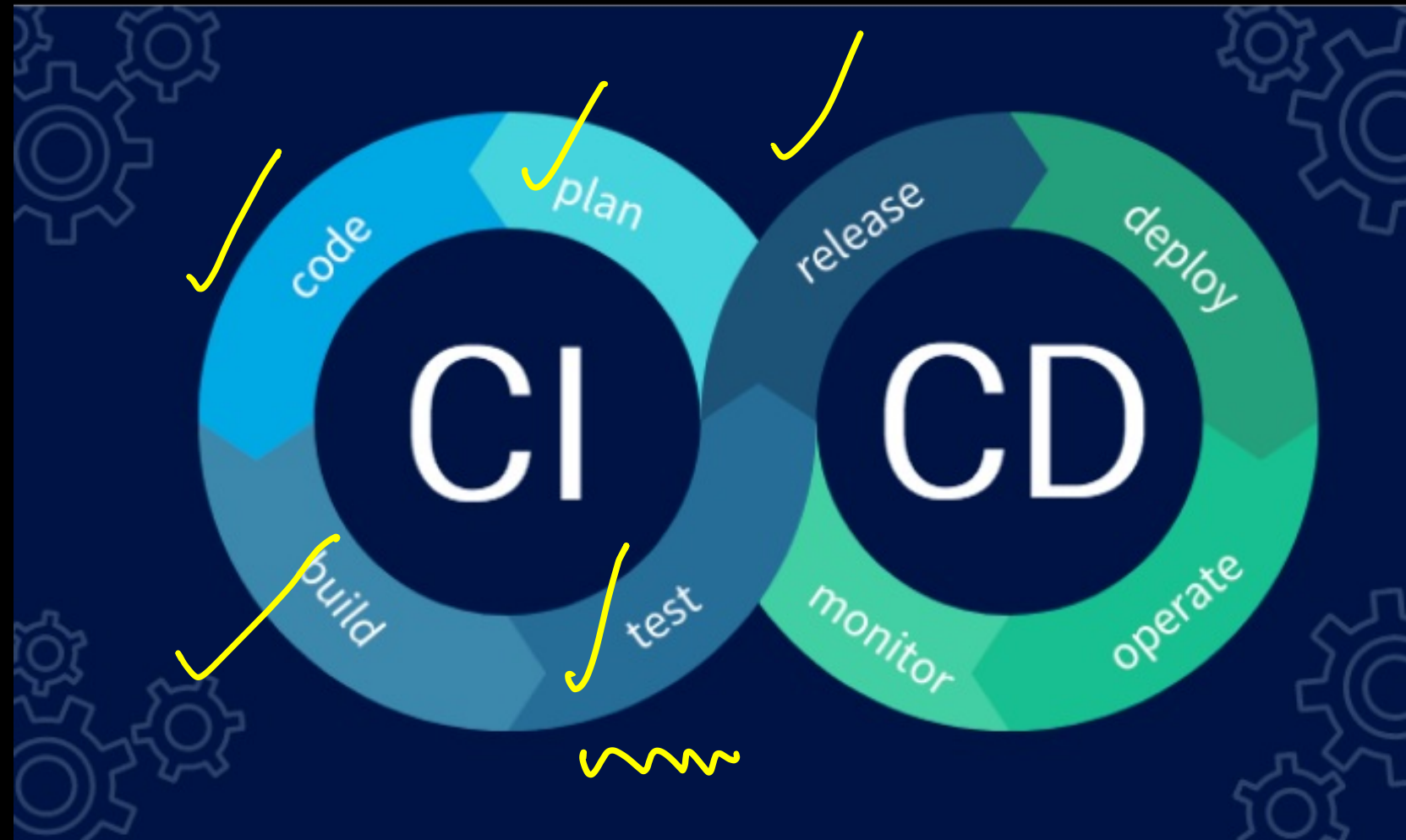
↳ Definition

↳ Unit test

↳ CI through github Actions

CI → Integrating code changes to main branch,
trigger automatic test cases that runs

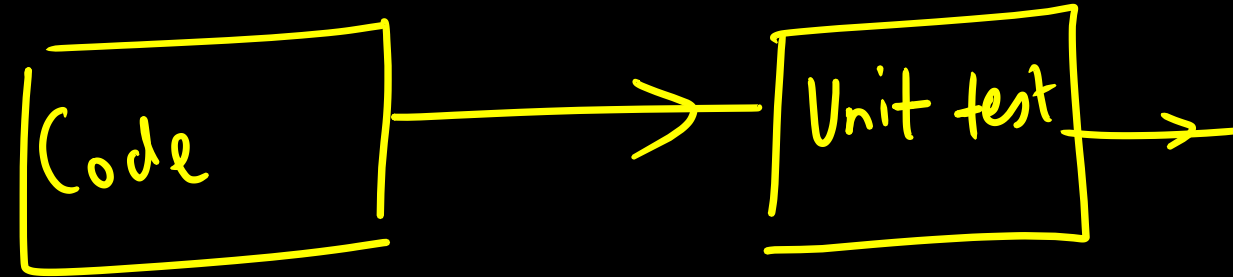




CI CD

CI
↓
CD
↓

Unit test



Pytest = Python testing framework

```
@pytest.fixture
def client():
    return app.test_client()

def test_home(client):
    resp = client.get("/ping/Sh")
    assert resp.status_code == 200
```

def test_user(client):

→

→

Pytest inject this client
into any test that
ask for it

Create a virtual browser.

Simulate HTTP request (GET, Post)

Status code

[200 → Success
404 → Error

YAML

1. name

2. on

→ push ✓

→ pull-request ✓

3. jobs

→

→

name: Python Package using Conda ✓

on: [push]

jobs:

build-linux: ✓

runs-on: ubuntu-latest ✓

strategy:

max-parallel: 5

mac
windows

✓ steps:

- uses: actions/checkout@v4

- name: Set up Python 3.10 ✓

uses: actions/setup-python@v3

with:

python-version: '3.10'

- name: Run test
run: pytest ✓

→ Break until 22:20 PM

Push
to main branch

VIM

Python 3.12.12

Ubuntu-latest

name: Run Python_test_for_flask_app

on:
push:
branches:
- main

jobs:

build:

runs-on: ubuntu-latest

steps:

- uses: actions/checkout@v2

- name: Install Python 3

uses: actions/setup-python@v1

with:

python-version: 3.12.12

- name: Install dependencies

run: |

python -m pip install --upgrade pip

pip install pytest

pip install -r requirements.txt

- name: Run tests with pytest

run: pytest

Actions

New workflow

All workflows

Run Python_test_for_flask_app

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

Filter workflow runs

1 workflow run

Event

Status

Branch

Acto

Added the requirements.txt and yaml file

Run Python_test_for_flask_app #1: Commit [95a4a8c](#) pushed by [153079019shariq](#)

main

now

In progress

Repository
- github / workflows

